

EXPERIMENT - 1

LINUX COMMANDS

AIM:

To Study various network commands used in linux and windows.

1. CODE:

arp -a

OUTPUT:

```
(172.16.11.109) at 00:27:0e:13:f7:8c [eth0] on enp3s0
(172.16.8.149) at 00:27:0e:13:f3:fe [eth0] on enp3s0
(172.16.8.186) at 7c:51:58:38:e6:24 [eth0] on enp3s0
(172.16.9.212) at 7c:51:58:38:da:a3 [eth0] on enp3s0
```

2. CODE:

hostname

OUTPUT:

localhost.localdomain

3. CODE:

netstat -r

OUTPUT:

kernel IP routing table

Destination	Gateway	Genmask	Flags	Uss/ Window	irtt	Interface
default	gateway	0.0.0.0	UG	0 0	0	enp3s0
172.16.8.0	0.0.0.0	255.255.252.0	U	0 0	0	enp3s0

4. CODE:
nslookup www.facebook.com

OUTPUT:

Server: 172.16.8.1
Address: 172.16.8.1#53

Non-authoritative answer:

www.facebook.com
Canonical name = star-mini.clor.facebook.com
Name: star-mini.clor.facebook.com
Address: 57.144.56.1

5. CODE:
route

OUTPUT:

Kernel IP routing table

Destination	Gateway	Genmask	Flags	Metric	Ref	Use	Interface
default	gateway	0.0.0.0	UG	100	0	0	enp3s0
172.16.8.0	0.0.0.0	255.255.252.0	U	100	0	0	enp3s0

6. CODE:
ip address show

OUTPUT:

1:lo: inet 127.0.0.1/8 scope host lo
2: enp3s0: inet 172.16.8.127/22 brd 172.16.11.255
scope global enp3s0

7. ifconfig

OUTPUT:

enp3s0 : flags = 4163 <UP, BROADCAST, RUNNING MULTICAST>
mtu 1500 0

inet 172.16.11.207

netmask 255.255.252.0

broadcast 17.16.8.255

8. mtr - Serves as a network diagnostic tool

OUTPUT:

mtr google.com

Local Host: local domain()

Host

Packets

loss %	snt	lost	Avg	Root	wtc	stddev
0.01 %	17.8	0.1	0.1	0.0	0 %	0.0

9. tcpdump - designed to capture and display packets

Output:

tcpdump - D

1. enp3s0 (up, Running)

2. any [pseudo: device that captures on all interfaces] [up, running]

3. lo (up, Running, Loopback)

4. wlp 370

5. Bluetooth 0 (Bluetooth adapter number 0)

10. nbtstat -s

OUTPUT:

NetBIOS Names Resolution and Registration Statistics

Resolved by Broadcast = 0

Resolved by Name Server = 0

Registered by Broadcast = 24

Registered by Name Server = 0

1. pathping -4 172.16.97.153

OUTPUT:

over a max. of 30 hops:

0 Sg 007 [172.16.97.153]

1 Sg007 [172.16.97.153]

Computing Statistics of 25 seconds

Hop	RTT	Source to Here Host/sent = Pct	This Node / Link Lost/sent = Pct	Address
0			0/100 = 0% 0/100 = 0%	172.16.97.153 172.16.97.153
1	0ms	0/100 = 0%		

Trace Complete

Signature

RESULT:

Thus the network commands using
Linux and windows are successfully
executed and the output is verified.